

Reversal and Truncation

Teaching note

Basu (1997), *Journal of Accounting and Economics*, with Dietrich, Muller, and Riedl
(2007), *Review of Accounting Studies*

ACCTG 597A · Fall 2026 · Sam Bonsall · Penn State

Basu's test rests on an untested equality

"I test my hypotheses in a Beaver et al. (1980) 'reverse' regression, with earnings as the dependent variable, because OLS standard errors and test statistics are better specified when the leading variable is specified as independent and the lagging variable as dependent. I regress annual earnings on current annual returns."

Basu (1997, p. 11)

The design "assumes that, in the absence of conservatism, earnings will respond identically to both 'good' and 'bad' news" (DMR, p. 97).

- Direction and sign partition are two choices about one variable, R_{it} ; each adds a bias term of unknown sign to β_1 .

■ Basu's β_1 is the difference of two subsample slopes

Basu Table 1 (p. 13): price-deflated earnings on the annual return, with an intercept and slope dummy for negative returns.

$$X_{it}/P_{it-1} = \alpha_0 + \alpha_1 DR_{it} + \beta_0 R_{it} + \beta_1 R_{it} DR_{it}, \quad DR_{it} = 1 \text{ if } R_{it} < 0$$

Basu Table 1 Panel A: $\beta_0 = 0.059$, $\beta_1 = 0.216$ ($t = 20.66$); the bad-news slope is $(0.216 + 0.059)/0.059 = 4.66$ times the good-news slope.

DMR eq. (1.1), p. 100, no conservatism: $R_{it} = \beta X_{it} + \lambda \eta_{it}$ with $\beta = \lambda = 1$ and $\sigma_{X,\eta} = 0$; the Basu regression is the pair

$$X_{it} = \delta_0 R_{it} + \xi_0 \mid R_{it} \geq 0 \quad (\text{DMR 1.5a}),$$

$$X_{it} = \delta_1 R_{it} + \xi_1 \mid R_{it} < 0 \quad (\text{DMR 1.5b})$$

■ Hence $\beta_1 = \delta_1 - \delta_0$ (DMR fn. 14, p. 102): two slopes estimated on samples formed by the sign of the regressor.

Reversal replaces β with $\beta\sigma_X^2/\sigma_R^2$

From DMR (1.1) with $\sigma_{X,\eta} = 0$: $\sigma_{R,X} = \beta\sigma_X^2$ and $\sigma_R^2 = \beta^2\sigma_X^2 + \lambda^2\sigma_\eta^2$ (p. 100).

$$\text{forward, DMR (1.2): } \text{plim } \hat{\beta}' = \frac{\sigma_{R,X}}{\sigma_X^2} = \beta \qquad \text{reverse, DMR (1.3): } \text{plim } \hat{\delta} = \frac{\sigma_{R,X}}{\sigma_R^2} = \frac{\beta\sigma_X^2}{\beta^2\sigma_X^2 + \lambda^2\sigma_\eta^2} = \beta \left[\frac{\sigma_X^2}{\sigma_R^2} \right] \quad (\text{DMR 1.4})$$

$$\frac{1}{\text{plim } \hat{\delta}} = \beta + \frac{\lambda^2\sigma_\eta^2}{\beta\sigma_X^2} \geq \beta (= 1), \qquad R^2 = \text{plim } \hat{\delta} \cdot \text{plim } \hat{\beta}'$$

SVR bias: $\beta\sigma_X^2/\sigma_R^2 - \beta$ $\hat{\delta}$ recovers $1/\beta$ only if $\hat{\delta}^2 = R^2\sigma_X^2/\sigma_R^2$
the slope falls toward zero and $R^2 = 1$ Basu fn. 7 (p. 11) is DMR
the bias toward $-\beta$ as σ_η^2 DMR fn. 13; Week 1, BLR (1.4) squared (DMR fn. 12)
grows (DMR p. 102) Property (3), $\gamma\delta = R^2$

- The reverse slope is a forward slope scaled by a variance ratio, not a speed of recognition (DMR p. 101).

A split on returns truncates the error

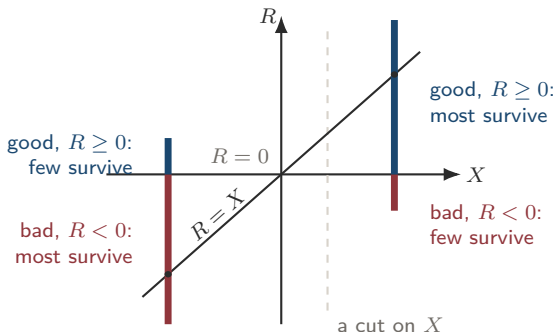
Structural line $R = X + \eta$; η is vertical at each X (DMR Fig. 2, p. 103).

Bad news, $R < 0$: at high X only very negative η survive; at low X most do.

Good news, $R \geq 0$: the same tilt, seen from above the cut.

A cut on X (dashed) truncates no η range.

- Sorting on R , which contains η , pairs high X with low η in both halves: both conditional covariances are negative.



Each subsample slope carries two bias terms

For either half j (good: $R \geq 0$; bad: $R < 0$), $\text{plim } \hat{\delta}_j = [\sigma_{R,X}/\sigma_R^2 \mid j]$ (DMR 1.6a, b). Substitute $\sigma_{R,X} = \beta\sigma_X^2 + \lambda\sigma_{X,\eta}$, which holds pointwise and so within any subsample:

$$\text{plim } \hat{\delta}_j = \beta \left[\frac{\sigma_X^2}{\sigma_R^2} \mid j \right] + \lambda \left[\frac{\sigma_{X,\eta}}{\sigma_R^2} \mid j \right] \quad (\text{DMR 1.7a, b})$$

DMR's maintained assumption (1.8) is $\text{plim } \hat{\delta}_0 = \text{plim } \hat{\delta}_1$; subtracting the two lines of (1.7):

$$\text{plim } \hat{\beta}_1 = \beta \left\{ \left[\frac{\sigma_X^2}{\sigma_R^2} \mid R < 0 \right] - \left[\frac{\sigma_X^2}{\sigma_R^2} \mid R \geq 0 \right] \right\} + \lambda \left\{ \left[\frac{\sigma_{X,\eta}}{\sigma_R^2} \mid R < 0 \right] - \left[\frac{\sigma_{X,\eta}}{\sigma_R^2} \mid R \geq 0 \right] \right\}$$

Navy: truncated variance ratios. Maroon: truncated covariances, negative in both halves.

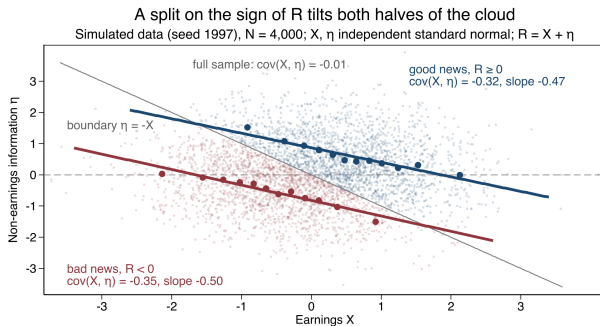
- Neither term is a timeliness parameter, and “In general, the equality of (1.9) will not hold” (DMR p. 104).

$\text{cov}(X, \eta)$ turns negative in both halves

X, η independent standard normal,
 $R = X + \eta$; bad news is $\eta < -X$.

Within-half slopes of η on X :
-0.47 (good), -0.50 (bad);
covariances -0.32 and -0.35,
closed form $-1/\pi = -0.318$.

■ This covariance is the
numerator of the **ST term** in
DMR (1.7): zero overall,
negative in both halves.



`basu_asymmetric_timeliness_truncation.do`; dots
are binned means.

The decomposition holds exactly in sample

```
set obs 50000
gen X = 0.17*rnnormal()
gen eta = 0.497*(rchi2(2)-2)/2
gen R = X + eta
gen DR = (R < 0)
gen RDR = R*DR
reg R X // b, tau
reg X R if R >= 0 // delta_0
reg X R if R < 0 // delta_1
// SVR_j = b*var(X|j)/var(R|j)-b
// ST_j = cov(X,tau|j)/var(R|j)
reg X DR R RDR // beta1
```

	Symmetric normal		Right-skewed η	
	Good, $R \geq 0$	Bad, $R < 0$	Good, $R \geq 0$	Bad, $R < 0$
Forward slope b	1.009	1.009	0.978	0.978
SVR_j	-0.745	-0.736	-0.860	-0.273
ST_j	-0.161	-0.170	-0.120	-0.227
$b + SVR_j + ST_j$	0.103	0.102	-0.002	0.478
Basu slope $\hat{\delta}_j$	0.103	0.102	-0.002	0.478
$cov(X, \eta j)$	-0.016	-0.017	-0.030	-0.009
R_j^2	0.041	0.039	0.000	0.317
$\hat{\beta}_1(t)$	-0.001 (-0.2)		0.480 (95.6)	

Seed 1997, N = 50,000, DMR Table 1 moments;
basu_asymmetric_timeliness_identity.do.

- Each Basu slope equals $b + SVR_j + ST_j$ to six decimals; with right-skewed η , $\hat{\beta}_1 = 0.480$ ($t = 95.6$) under the null.

Three conditions make the two biases cancel

DMR (pp. 104–105), $\beta = \lambda = 1$: the slopes are equal under (i) symmetric X ; (ii) symmetric, homoskedastic, mean-zero η ; (iii) a split at the mean of R .

Sufficient, not necessary: under joint normality $E[X | R]$ is linear, so the slopes agree whatever the split (slide 8; design F, slide 11).

- Then the truncated errors are “mirror images of each other” (p. 110); bias in β_1 requires a departure from normality.

	Full sample	Good, $R \geq 0$	Bad, $R < 0$
Full-sample forward slope $\hat{b}$	1.000	1.000	1.000
Variance ratio s_X^2/s_R^2	0.429	1.249	1.249
ST term $s_{X\tau}/s_R^2$	0.000	-0.436	-0.436
Basu slope $\hat{\delta} = \hat{b} \times \text{ratio} + \text{ST}$	0.429	0.813	0.813
Subsample forward slope $\hat{b}_j$ ($\hat{\delta}_j = \hat{b}_j \times \text{ratio}_j$)	1.000	0.651	0.651

DMR Fig. 3 Panel A recreated (pp. 105–110), twelve observations: each half’s slope is biased, 0.813 against 1, yet equal.

One violated condition suffices to sign β_1

“When one of these conditions is not met, biased test statistics obtain that are consistent with earnings being “conservative” or “aggressive”—even when earnings are purely transitory.”

Dietrich, Muller, and Riedl (2007, p. 105)

Twelve observations, $R = X + \eta$	Good $\hat{\delta}_0$	Bad $\hat{\delta}_1$	$\hat{\beta}_1 = \hat{\delta}_1 - \hat{\delta}_0$
A: X symmetric, η homoskedastic	0.813	0.813	0.000
B: variance of η rising in X	0.038	1.917	1.879
C: X left-skewed	0.281	0.602	0.321
Mirror of C: X right-skewed, as per-share earnings (DMR fn. 18)	0.602	0.281	-0.321

Recreated from DMR Fig. 3 Panels A to C (pp. 105–111), twelve observations each.

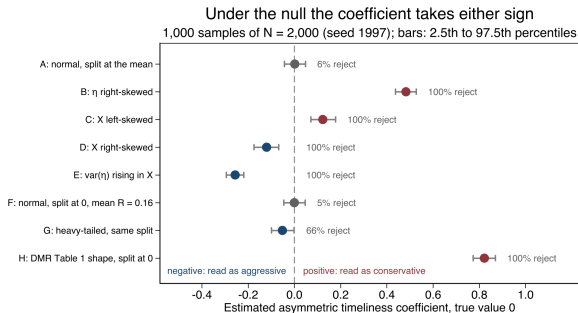
- The mirror row negates X and η , swapping the halves and flipping $\hat{\beta}_1$ exactly: the violated condition sets the sign.

Under the null, β_1 takes either sign

Eight designs with DMR Table 1 moments; 1,000 samples of $N = 2,000$; true $\beta_1 = 0$.

Normal data (A, F) center at zero, split at or off the mean; the other six reject in 66 to 100 percent of samples, means -0.256 (E) to 0.821 (H).

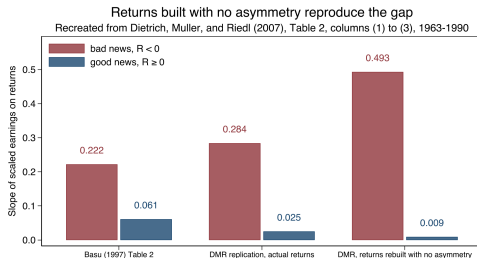
- Design E reverses the sign of DMR Fig. 3 Panel B: the sign depends on moments of η the researcher does not observe.



`basu_asymmetric_timeliness_montecarlo.do`

A placebo return reproduces Basu's asymmetry

DMR rebuild each return from the forward regression and another firm's residual, so it carries no firm-specific asymmetry (p. 113). DMR Table 2 Panels D and E (p. 115), actual returns: bad $0.852 - 0.055 - 0.513 = 0.284$, good $0.852 - 0.768 - 0.059 = 0.025$; ST dominates the bad-news slope, SVR the good-news slope.



`basu_asymmetric_timeliness_table2.do`;
Basu's column is Basu Table 2 (XE, 34,266 firm-years).

- Like with like: the placebo $\hat{\beta}_1 = 0.493 - 0.009 = 0.484$ exceeds Basu's own, 0.216 (Basu Table 1 Panel A) and 0.161 (Basu Table 2).

Other sign partitions inherit the same bias

Maroon rows split on a variable that contains the error of the structural relation; navy rows are the contrasts.

Design	Partition variable (contains the error?)	What the split produces
Basu Table 1 (p. 13)	sign of R , the reverse regressor (yes: η)	SVR and ST; 0.275 versus 0.059
DMR Table 3, B and C (p. 117)	market-to-book level, no news (yes: skewed deflated price)	0.193 versus 0.006; 0.123 versus 0.009
Basu Table 3, B (p. 21)	sign of the lagged change, its own regressor (plausibly)	-0.692 versus -0.040
A split on X (DMR Fig. 2, p. 103)	the reverse regressand (no: η untouched)	SVR term only
Basu Table 4 (p. 25)	sign of the earnings change, the forward regressor (no)	no truncation term; 0.114 ($t = 7.67$)
Week 1: BLR (1987)	groups formed on the regressand (yes)	grouping bias

The Basu Table 3 row is a question, not a finding: a change split on its own sign mean-reverts, so part of -0.692 may be reversal of a transitory component.

- Whenever the partition variable contains the structural error, the difference between two subsample slopes is not a parameter of the accounting process.

The paper names truncation only for survival

Basu fn. 14 (p. 22) names truncation only for survivorship, “Maddala (1991) suggests correcting for this by using a truncated regression model,” not for the split on R .

“Further, there does not appear to be an operative solution to address these biases, as they originate in the asymmetric timeliness specification itself.”

Dietrich, Muller, and Riedl (2007, p. 120)

A skeptic would note that Ball and Kothari (2006; DMR fn. 16) derive “substantially equivalent” conditions for a well-specified test under the null.

- DMR’s reply (p. 105, fn. 16): with DMR Table 1’s tails, those conditions are unlikely to hold; slides 11 and 12 show the size of the bias.

Summary

- Reversing a structural regression and partitioning on the sign of its regressand changes what the slope estimates: within each subsample, the Basu slope equals the forward slope scaled by a truncated variance ratio plus a truncated covariance term, and neither component is a timeliness parameter.
- The interaction coefficient is the difference of two such slopes, so its sign and magnitude under the null depend on the skewness of scaled earnings, the distribution of non-earnings information, and the location of the split relative to the mean of returns, none of which the researcher observes.
- Because the biases originate in the specification rather than in the estimator, an asymmetric timeliness coefficient of either sign appears consistent with earnings that are not conservative, which suggests that the design cannot by itself detect conservatism.

Basu (1997) · Dietrich, Muller, and Riedl (2007) · Beaver, Lambert, and Ryan (1987) · github.com/sbb151/acctg597a-research-design